

2.12 Fluid Compartments and Solutes

▼ Which cation is found predominantly in cells?

- K^+

▼ Which cation is found predominantly outside cells?

- Na^+

▼ Which anion is found predominantly in cells?

- Organic phosphates

▼ Which anion is found predominantly outside cells?

- Chloride ions
- These help to balance the effects of K^+ inside cells

▼ Is the inside or the outside of a cell more acidic?

- Inside.

▼ What is osmolarity?

- A measure of the **concentration of all solute particles in a solution**
- i.e a cumulative concentration of all the solutes.

▼ Why is there no osmosis between interstitial fluid and cells?

- The **osmolarity inside and outside of cells is equal.**
- This means there is no net movement of water in and out of cells.

▼ Table showing how osmolarity is measured.

	Concentration (mol/L)	Osmoles/L
	1	2 (Na^+ , Cl^-)
	1	3 (Ca^{2+} , $2Cl^-$)
e	2	2 (glucose)
arity		7

- The key principle is that each ionic compound is split into its constituents and each constituent is counted as an **osmole**.

▼ What is diffusion?

- The spontaneous movement of a solute **down a concentration gradient until** the solute molecules **reach an equilibrium**.

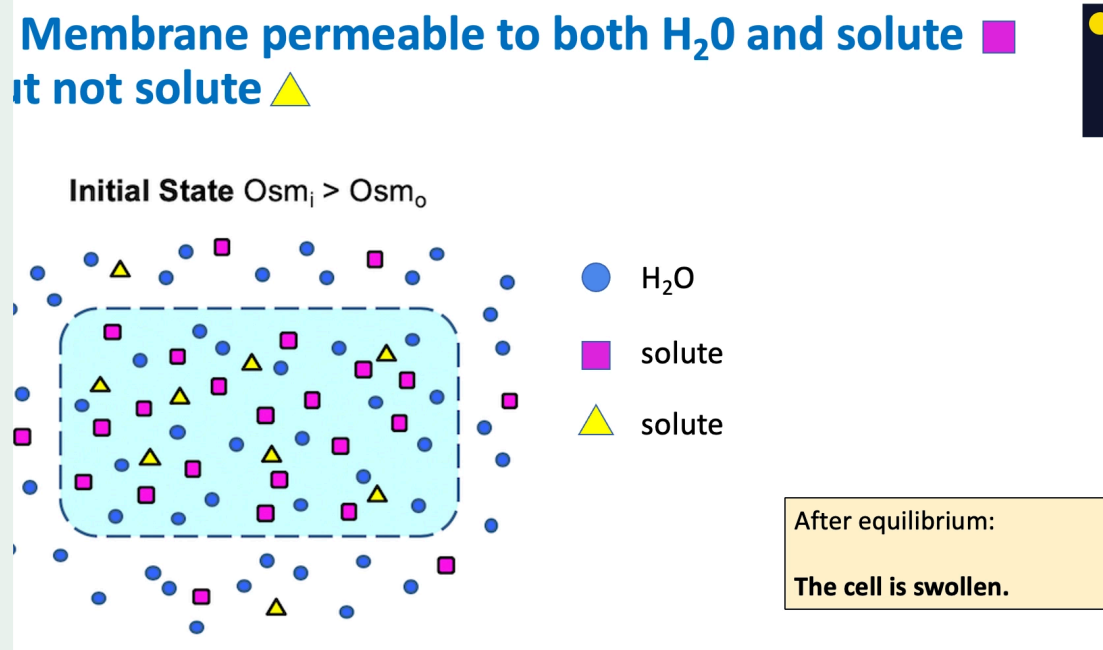
▼ What is osmosis?

- The **movement of water towards an area of higher osmolarity** across a partially permeable membrane

▼ What is an osmole?

- The number of moles of a solute that contribute to the solution's osmotic pressure.
 - i.e how many moles of the solute are present

▼ Describe what happens in the below scenario.



▼ Answer

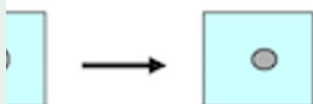
- **Purple solute (P) move out of the cell** down the concentration gradient.
 - **No net movement of water is caused by the purple solute**
- Yellow solute (Y) cannot move in or out of the cell as membrane is impermeable to it, so **water moves into the cell as solute Y causes high intracellular osmolarity**
- At the **equilibrium (resting) state, the cell is swollen.**

- ▼ Why is osmolarity on its own an unreliable way to judge whether a cell bursts or swells?
- Because **osmolarity can guide the direction which water** will move but permeability of the cell membrane to solutes is also very important
 - i.e a cell membrane could be impermeable to an intracellular solute, meaning that water will move inside in the equilibrium state to balance osmolarity
 - in contrast, if the cell membrane was permeable to the solute, it would move outwards on its own, and not as much water would need to move into the cell to balance osmolarity
 - Hence the **osmolarity of two identical cells can be equal**, but different amounts of water go in/out of it **depending on its permeability to solutes**

- ▼ When can a cell rupture?
- When the difference in osmolarity between inside and outside of the cell is so high that so much water moves in that can cause rupturing.
 - i.e **very high intracellular osmolarity**

- ▼ What is tonicity?
- Tonicity describes the **osmolarity of a solution's impermeant solutes** present in the solution
 - This hence helps us to understand which direction water needs to move in, to balance this osmolarity out

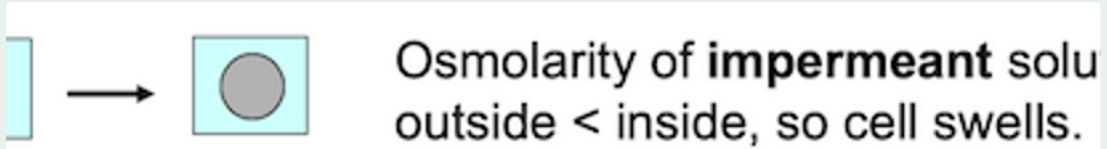
- ▼ What is a hypertonic solution?
- One which has a **greater osmolarity of impermeant solutes than the inside of the cells** around it
 - This leads to **water having to move out of the cell** to balance the osmolarities, **so the cell shrinks**



Osmolarity of **impermeant solutes** outside > inside, so cell shrinks.

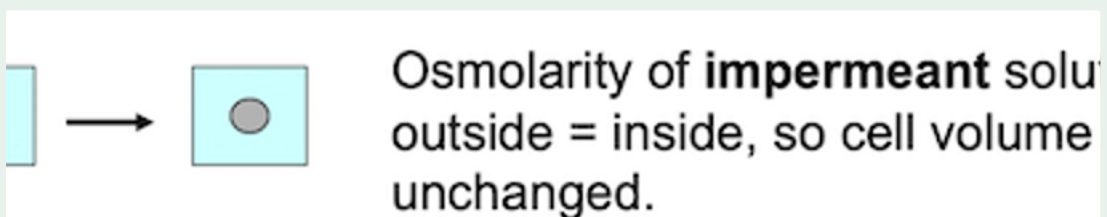
▼ What is a hypotonic solution?

- One which has a lower osmolarity than the inside of the cells around it
 - This leads to **water having to move into the cell** to balance the osmolarities, **so the cell swells**



▼ What is an isotonic solution?

- One which has an **identical osmolarity of impermeant solutes to the inside of the cells around it**
 - Water does not need to move in any particular direction to balance out the osmolarities, because all the **solute can move themselves**
- Basically, water only moves when the solutes cannot move across their concentration gradients



▼ What mechanism do cells have which allows them to not burst?

- **Na⁺/K⁺ ATPase**
- This constantly pumps Na⁺ out of the cell, increasing extracellular osmolarity and allowing water to not accumulate within the cell, but H₂O to follow Na⁺

▼ What is an alternative name for the sodium-potassium pump?

- Na⁺/K⁺ ATPase.

▼ Why does the sodium-potassium pump not function in organs outside the body?

- It **cannot function below 15°C**
- There is **no ATP being produced in ischaemic conditions** so it cannot function

▼ What is University of Wisconsin (UW) solution used for?

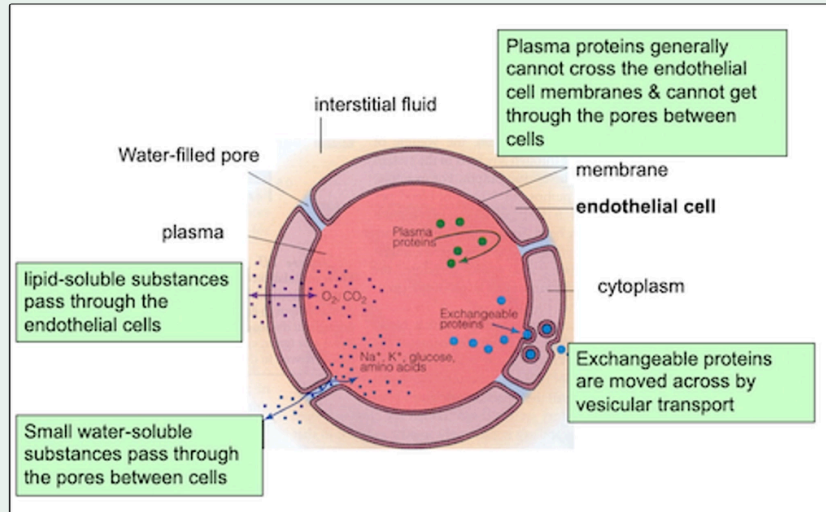
- To reduce cell-swelling in tissues of organs that are preserved in it.

▼ What three properties of UW solution help it to reduce cell swelling?

- **Lack of Na^+/Cl^-** which prevents swelling
 - Otherwise, these ions would move down the conc. gradient into the tissues, and hence drag water into the tissues, causing swelling
- Presence of **extracellular impermeable solutes**
 - This increases extracellular osmolarity, meaning water is drawn out of tissues
- Presence of **starch** to contribute to and **increase colloidal osmotic pressure**
- **Antioxidants to protect tissue from damage** via reactive oxygen species.

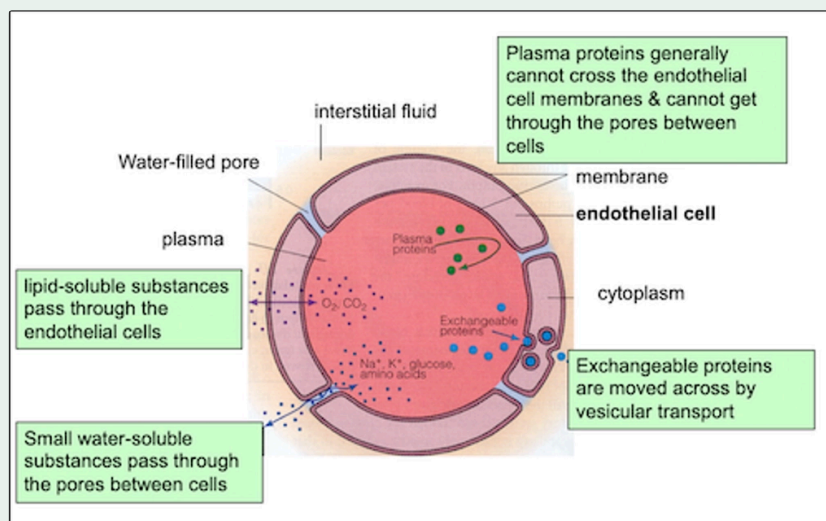
▼ What property of the endothelial layer allow them to transfer blood plasma to the interstitial space?

- **Leaks or pores (fenestrae)** in between the endothelial cells.



▼ What is colloidal osmotic pressure (COP)?

- Also known as **oncotic pressure**, COP is a type of **osmotic pressure induced by the plasma proteins**, notably **albumin** in a blood vessel's plasma
- This pressure is caused by the **plasma proteins within blood vessels being unable to move out of the blood vessels**, so they build up and cause pressure in the form of COP
 - It acts to **pull water into blood vessels**, because the **accumulation of plasma proteins causes high intravascular osmolarity**. Water moves into vessels to **balance the high osmolarity found within the vessel**



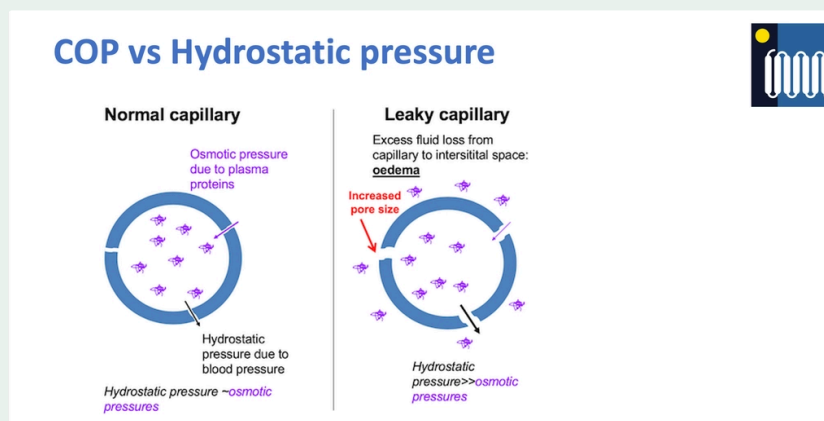
▼ What is hydrostatic pressure?

- The **"pushing" force exerted by any fluid** as a result of it moving through its concordant vessel

▼ Can plasma proteins fit through the endothelial wall?

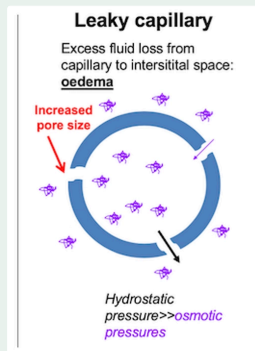
- No.

- ▼ How do the colloidal osmotic pressure and hydrostatic pressure balance to determine the movement of molecules in and out of the endothelium?
- **Colloidal osmotic pressure** can be induced by **high amounts of plasma proteins** accumulating in the lumen
 - **Hydrostatic pressure** is induced by the **movement of the blood** through the blood vessel
 - The balance of colloidal/hydrostatic pressure can cause **soluble ions and even plasma proteins** to be **pushed through** the endothelial wall **between the pores**.
 - i.e Higher hydrostatic pressure compared to colloidal osmotic pressure can lead to fluid being more readily pushed out to balance things out
 - The aim is to have an equilibrium where **colloidal osmotic pressure is equal to hydrostatic pressure**
 - **Hydrostatic pressure** > COP = fluid moving **out of** blood vessel
 - **COP** > Hydrostatic pressure = fluid moving **into** blood vessel



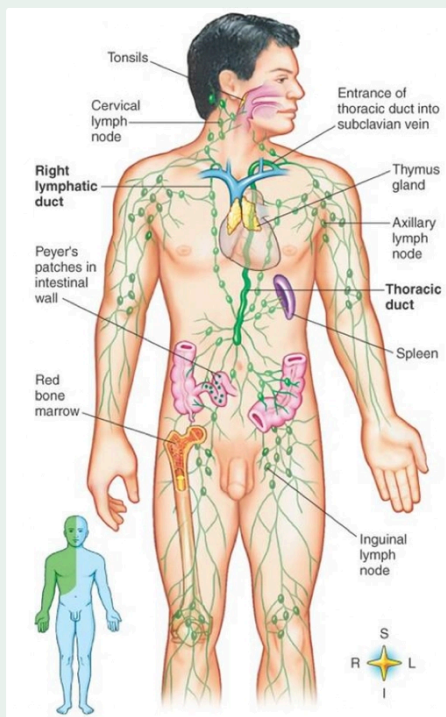
▼ What happens to the blood vessels in oedema (tissue swelling)?

- Excess **fluid leaks from the lumen of blood vessels into the interstitial space** which causes swelling
- This can be **caused by extra leaky capillaries**, due to the imbalance of colloidal osmotic pressure and hydrostatic pressure
 - **Hydrostatic pressure > COP**



▼ What is the function of the lymphatic capillaries?

- To **collect interstitial fluid that has leaked from the blood vessels** as a result of **hydrostatic pressure exceeding oncotic pressure**
- It returns this fluid back to the blood capillaries via the **lymphatic ducts of the subclavian region, or the lymph nodes**



▼ How can the lymphatic capillaries help to prevent oedema?

- They can **recollect plasma that is lost** from the blood vessels to the interstitial fluid and **prevent swelling of tissues**
- They **uptake this interstitial fluid** and return it to the capillaries for **circulation**.

